

CELL BIOLOGY LECTURE 1: INTRODUCTION TO CELLS

1. Cell Theory

The cell theory has three main principles:

- All living organisms are composed of one or more cells.
- The cell is the basic unit of structure and organization in organisms.
- All cells arise from pre-existing cells.

2. Prokaryotic vs Eukaryotic Cells

Prokaryotic cells:

- Lack a membrane-bound nucleus
- Lack membrane-bound organelles
- Smaller (0.1-5 micrometers)
- Examples: bacteria, archaea
- DNA is circular and in the nucleoid region
- Have ribosomes (70S)

Eukaryotic cells:

- Have a membrane-bound nucleus
- Have membrane-bound organelles
- Larger (10-100 micrometers)
- Examples: plants, animals, fungi, protists
- DNA is linear and in the nucleus
- Have ribosomes (80S)

3. Cell Organelles and Their Functions

Nucleus: Contains genetic material (DNA), controls cell activities

Mitochondria: Site of ATP production through cellular respiration

Endoplasmic Reticulum (ER):

- Rough ER: Has ribosomes, involved in protein synthesis
- Smooth ER: No ribosomes, involved in lipid synthesis

Golgi Apparatus: Modifies, sorts, and packages proteins

Lysosomes: Contains digestive enzymes, breaks down waste

Ribosomes: Protein synthesis

Chloroplasts: Site of photosynthesis (plants only)

Cell Membrane: Phospholipid bilayer, controls what enters and exits

4. Cell Membrane Structure

The cell membrane is a phospholipid bilayer with embedded proteins.

Key components:

- Phospholipids: hydrophilic head, hydrophobic tail
- Channel proteins: allow specific molecules to pass
- Carrier proteins: transport molecules across membrane
- Receptor proteins: receive chemical signals
- Cholesterol: provides membrane stability (animal cells only)

5. Transport Across Cell Membrane

Passive Transport (no energy required):

- Simple diffusion: molecules move from high to low concentration
- Facilitated diffusion: uses channel/carrier proteins
- Osmosis: diffusion of water across membrane

Active Transport (energy required):

- Uses ATP to move molecules against concentration gradient
- Sodium-potassium pump is a key example

6. Key Terms and Definitions

Homeostasis: The ability to maintain a stable internal environment